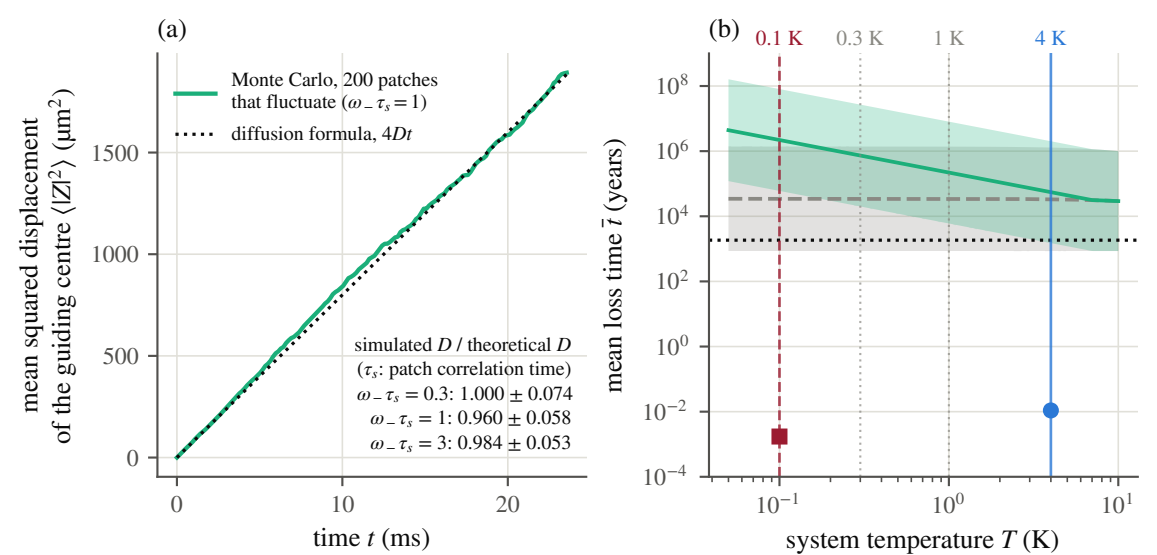
(a)
Monte Carlo, 200 patches that fluctuate ($\omega_- \tau_s = 1$)
diffusion formula, $4Dt$
simulated D / theoretical D
(τ_s: patch correlation time)
$\omega_- \tau_s = 0.3$: 1.000 ± 0.074
$\omega_- \tau_s = 1$: 0.960 ± 0.058
$\omega_- \tau_s = 3$: 0.984 ± 0.053
mean squared displacement of the guiding centre $\langle |Z|^2 \rangle$ (μm^2)
time t (ms)
(b)
0.1 K
0.3 K
1 K
4 K
mean loss time $\bar{t}$ (years)
system temperature T (K)
Panel (b), surface noise: the band is the possible range and the line its central value
A: patch noise independent of temperature
B: patch noise freezes out on cooling
minimum compatible with BASE
transit model, 4 K
transit model, 0.1 K